

Tianchu Zeng

PhD Candidate, Electrical & Computer Engineering · National University of Singapore
Singapore · +65 8790 5814 · tianchuzeng@gmail.com

ORCID: [0009-0009-0773-4502](https://orcid.org/0009-0009-0773-4502) · [Google Scholar](https://scholar.google.com/citations?user=...) · [GitHub](https://github.com/tianchuzeng) · [LinkedIn](https://www.linkedin.com/in/tianchuzeng) · [Website](https://tianchuzeng.github.io)

RESEARCH PROFILE

NUS PhD candidate in computational neuroscience and AI for science, developing methods for scalable brain modeling and statistically valid machine-learning evaluation. Developed DELSSOME, a deep-learning framework that accelerates biophysical brain-model fitting by 50–100×, enabling lifespan mapping of model-derived cortical excitation–inhibition (E/I) ratio in 12,005 participants. Co-developed SHARP, a statistical method for valid model comparisons under cross-validation, and co-led a meta-analysis of biomedical AI studies. Independently developed an AI-agent workflow for evidence extraction and statistical re-analysis. Co-first author of a study accepted for publication in Nature Methods.

EDUCATION

National University of Singapore — Singapore

Aug 2022 - Present (expected 2027)

Ph.D. in Electrical & Computer Engineering

- Advisors: Prof. BT Thomas Yeo and Prof. Thomas E. Nichols (University of Oxford)
- Thesis: Machine Learning in Brain Imaging

Tsinghua University — Beijing, China

Aug 2018 - Jun 2022

B.Eng. in Department of Automation

RESEARCH INTERESTS

Computational neuroscience · AI for brain science · scientific and brain foundation models · population-scale brain modeling · statistical methods for machine-learning evaluation · AI agents for scientific research

RESEARCH EXPERIENCE

DELSSOME — Scalable brain modeling and lifespan E/I trajectories

Aug 2022 - Present

Co-first-author study · Nature Methods, accepted

- Developed DELSSOME, a deep-learning framework that predicts fitting objectives from candidate parameters and empirical connectivity. Accelerated fitting-cost evaluation by 1,500–8,000× and overall fitting (including validation and testing) by 50–100× versus Euler-based methods across three biophysical brain models while preserving fitting accuracy in evaluated experiments; speedups exclude one-time data generation and training.
- Proposed an architecture redesign and rapidly completed experiments to strengthen the method.
- Co-led modeling of 12,005 participants across 14 datasets (ages 4.5–98.4), mapping model-derived cortical E/I trajectories: decline through development and adulthood followed by a late-life increase, with sex differences and a sensory–association gradient.

SHARP — Valid model comparisons under cross-validation

2024 - Present

Co-first-author preprint · Under review

- Co-developed SHARP, a statistical method that redesigns cross-validation to estimate variance and correlation for valid model comparisons. Benchmarked against 12 existing tests across 420 simulation scenarios, achieving the best balance of false-positive control and power in those scenarios.
- Co-led a PRISMA-guided meta-analysis of 184 biomedical AI studies across 30 fields and proposed the published-paper re-analysis; 97% used tests that ignored between-fold correlation. Among 68 re-analyzed studies, 69% (47/68) contained at least one comparison that lost significance after accounting for this correlation.
- Among re-analyzed studies whose abstract claims relied on invalid tests, 59% had at least one supporting comparison lose significance.
- Developed an AI-agent workflow based on SHARP's meta-analysis procedures, supporting evidence extraction and reproducible statistical re-analysis of published model comparisons.

Brain Foundation Models — Rigorous benchmarking for phenotypic prediction

2025 - Present

- Build and fine-tune brain foundation models; benchmark phenotypic prediction across foundation models and against classical machine-learning methods.

PUBLICATIONS

* denotes equal contribution and co-first authorship; + denotes co-corresponding authorship; author name in bold.

Research Articles

1. **Zeng, T.***, Tian, F.*, Zhang, S., Li, X., Tan, A. P., Larsen, B., ..., Yeo, B. T. T., & Alzheimer's Disease Neuroimaging Initiative (2026). Optimizing biophysical large-scale brain circuit models with deep neural networks. *Nature Methods*. Accepted for publication. [Preprint](#).
2. Zhang, S.*, Roemer-Cassiano, S. N.*, Chong, J. R.*, Tian, F., Li, Y., Wang, Z., **Zeng, T.**, Li, X., Deco, G., Venketasubramanian, N., Zhou, J. H., Chen, C. L.-H., Alzheimer's Disease Neuroimaging Initiative, Lai, M. K. P., Franzmeier, N.+, & Yeo, B. T. T.+ (2026). Excitation-Inhibition Imbalance Links Amyloid Pathophysiology to Cognition in Non-Demented Individuals. *Alzheimer's & Dementia*. Accepted for publication.
3. Qu, Y. L.+, Chen, J., Tam, A., Ooi, L. Q. R., Dhamala, E., Cocuzza, C. V., Zhang, S., **Zeng, T.**, ... & Holmes, A. J.+ (2025). Distinct brain network features predict internalizing and externalizing traits in children, adolescents and adults. *Nature Mental Health*, 3(3), 306-317.
4. Zhang, S.*, Larsen, B.*, Sydnor, V. J.*, **Zeng, T.**, An, L., Yan, X., ..., Satterthwaite, T. D.+, & Yeo, B. T. T.+ (2024). In vivo whole-cortex marker of excitation-inhibition ratio indexes cortical maturation and cognitive ability in youth. *Proceedings of the National Academy of Sciences*, 121(23), e2318641121.

Invited Research Briefings

5. **Zeng, T.**, & Yeo, B. T. T. (2026). Deep learning charts the excitation-inhibition balance of the brain throughout the lifespan. *Nature Methods*. Invited Research Briefing accompanying the DELSSOME study. Accepted.

Peer-Reviewed Conference Proceedings

6. Gui, N.*, **Zeng, T.***, Hu, J., & Zhang, Y. (2022). Visual-angle Attention Predictor: a multi-agent trajectory predictor based on variational auto-encoder. *22nd COTA International Conference of Transportation Professionals*, Changsha, China, Jul 8-11, 2022.
7. **Zeng, T.***, Gui, N.*, Hu, J., & Zhang, Y. (2022). Agents Separated Prediction Former: a novel multi-agent trajectory prediction model based on Transformer with 2D input. *22nd ASCE International Conference on Transportation and Development*, Seattle, USA, Jun 1-3, 2022.

Preprints

8. **Zeng, T.***, Li, H.*, Zhang, S.*, Tan, Y. Q., Tian, F., Orban, C., ..., Nichols, T. E.+, & Yeo, B. T. T.+ (2026). Spurious model comparisons are widespread in biomedical artificial intelligence. *bioRxiv* preprint. Under review. [DOI](#).

SCIENTIFIC SOFTWARE AND OPEN SOURCE

[DELSSOME](#) — Lead Developer

- Lead development and maintenance of the official implementation, including group- and individual-level pipelines, pretrained models, reproducible examples, documentation, and regression tests.

[SHARP](#) — Lead Developer

- Developed and publicly released a scikit-learn-compatible Python package for statistical model comparison under cross-validation.

SHARP Meta-analysis Agent Workflow — Sole Developer

- Independently developed a workflow for paper/code evidence extraction, deposited-result ingestion, reproducible re-analysis, and source-referenced reporting; preparing public release.
- Implemented evidence checks, selective second-agent review, handling of missing evidence, and regression tests for ten synthetic examples.

[ThomasYeoLab CBIG](#) — Research Software Contributor

- Contribute code, pull-request reviews, and maintenance to CBIG, an open-source neuroimaging software repository; authored and maintain Zeng2026_DELSSOME and reviewed the Zhang2024_pFIC pull request.

[admissions-ops](#) — Creator and Maintainer

- Built a local-first AI-agent workspace for admissions planning, with source-backed reports, deterministic validation, and required human review.

INVITED TALKS

1. Optimizing biophysical large-scale brain circuit models. *Asia-Pacific Computational and Cognitive Neuroscience Conference*, RIKEN, Japan, Jul 21-23, 2025. Invited talk.

CONFERENCE PRESENTATIONS AND POSTERS

1. Optimizing biophysical large-scale brain circuit models with deep neural networks. Organization for Human Brain Mapping, Bordeaux, France, Jun 14-18, 2026. Poster.
2. Optimizing biophysical large-scale brain circuit models with deep neural networks. NUHS Scientific and Innovation Summit, Singapore, Apr 1-2, 2026. Poster and Best Poster Award (Basic Science).
3. Optimizing biophysical large-scale brain circuit models with deep neural networks. AI4X, National University of Singapore, Jul 8-11, 2025. Poster.

AWARDS HONORS AND SCHOLARSHIPS

- Best Poster Award (Basic Science), NUHS Scientific and Innovation Summit, 2026
- NUS Research Scholarship, National University of Singapore, 2022-2026

ADDITIONAL RESEARCH EXPERIENCE

Brain Parcellation with Resting-State fMRI — National University of Singapore

Aug 2022 - Jun 2023

Supervisor: Prof. BT Thomas Yeo

- Developed mathematical and machine-learning methods for group- and individual-level brain parcellation from resting-state fMRI, with functional-connectivity analysis and cortical-surface visualization.

Traffic Trajectory Prediction with Graph Neural Networks — Tsinghua University

Jan 2020 - Jun 2022

Undergraduate Researcher · Advisors: Prof. Jianming Hu and Prof. Yi Zhang

- Built RNN-, GNN-, CVAE-, and Transformer-based models for multi-agent traffic trajectory prediction; constructed a simulated dataset for the Intelligent Winter Olympics project.
- Work produced two co-first-authored peer-reviewed conference papers and one granted Chinese patent (ZL 2021 1 1394192.5).

EEG Analysis for Fatigue Driving — Tsinghua University

Mar 2021 - Jun 2021

Undergraduate Researcher · Advisor: Prof. Xiaorong Gao

- Processed driving-scenario EEG data and designed and tuned a Wide & Deep network for driver-fatigue classification.

RESEARCH MENTORING AND SUPERVISION

- Supervise Hetu Li (PhD student, NUS) on SHARP and the accompanying biomedical AI meta-analysis.
- Supervise Wanyu Che (PhD student) and Zijie Ling (research assistant) at NUS on brain foundation-model development and benchmarking; guide model building and fine-tuning, review code, and support manuscript preparation.

TEACHING

- EE3731C Signal Processing, Teaching Assistant, National University of Singapore, AY2023-2024 Semester 1 and AY2024-2025 Semester 1
- EE2026 Digital Design, Teaching Assistant, National University of Singapore, AY2022-2023 Semester 2

ADDITIONAL TRAINING

- Cold Spring Harbor Asia Summer School on Computational and Cognitive Neuroscience, 2023

RESEARCH SKILLS

- **Programming and scientific computing:** Python, MATLAB, C/C++; PyTorch, JAX, scikit-learn, SciPy
- **Machine learning:** deep learning, foundation-model fine-tuning and evaluation
- **Statistics:** inference for cross-validation, hypothesis testing, simulation-based validation, power and confidence-interval evaluation; systematic review and meta-analysis
- **Neuroimaging:** resting-state fMRI, functional connectivity, brain parcellation, cortical-surface visualization, FreeSurfer/Freeview
- **Research engineering:** Git, GitHub Actions, Linux/bash, high-performance computing (PBS), conda, Docker; reproducible workflows and automated validation

PROFESSIONAL MEMBERSHIPS

- Organization for Human Brain Mapping, Student member, 2024 and 2026